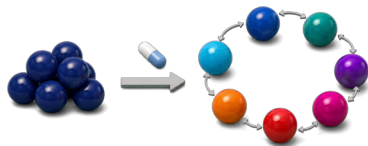


Drug-tolerant cancer cells cycle between states, quantifiable as a rising $\max H_1$

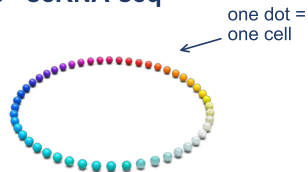
(a) drug pressure induces cycling → (b) scRNA-seq sees it as a loop → (c) $\max H_1$ rises across 4 cohorts

a · biology



Drug pressure makes cancer cells cycle between states

b · scRNA-seq

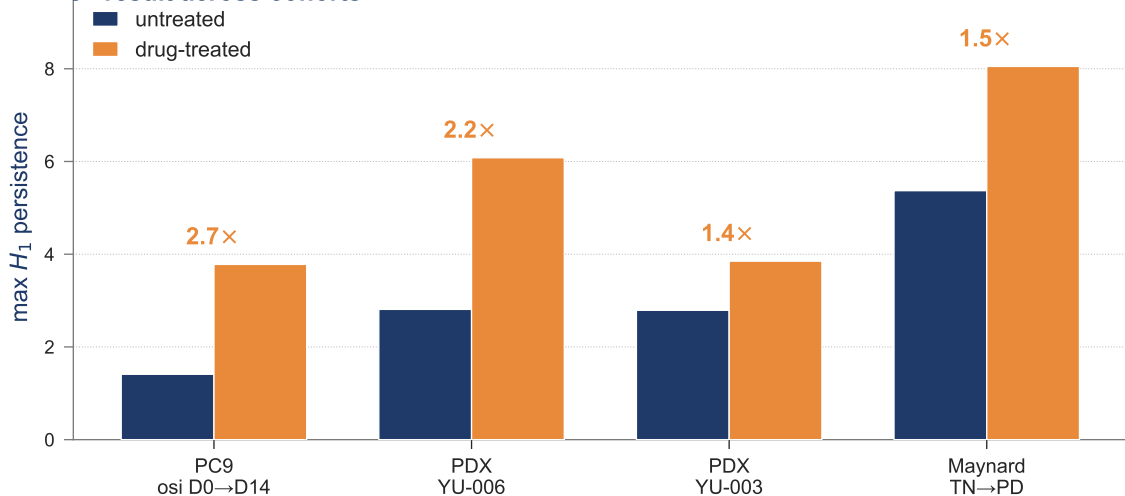


scRNA-seq sees the cycle as a closed loop



quantify with persistent homology

c · result across cohorts



Drug treatment raises $\max H_1$ across cell line, PDX, and patient cohorts